Assignment 3

Analyze a dataset of climate-related tweets collected from different countries.
Dataset: `Climate_Tweets_Sample.csv`

In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import sklearn
```

Part A – Cleaning & Sentiment Overview

Dataset

In [2]:
```
df = pd.read_csv('https://raw.githubusercontent.com/TheChroniclerr/CP321/refs/heads/main/a3/assets/Climate_Tweets_Sample.csv')
df.head()
```

Out[2]:

	tweet_id	date	country	sentiment	keyword	text	retweet_count	like_count
0	2803826677020224	2023-11-01	Denmark	negative	air quality	Breaking: concerns grow over air quality; data...	80.0	92.0
1	1119349365897942	2022-11-16	Spain	negative	air quality	Reminder: concerns grow over air quality; deta...	14.0	125.0
2	3605591970800719	2023-01-20	France	neutral	green transition	FYI: discussion continues on green transition;...	78.0	109.0
3	4546715437012377	2023-09-10	Saudi Arabia	negative	green transition	Opinion: concerns grow over green transition; ...	75.0	63.0
4	9630049315224554	2022-09-26	Ireland	negative	global warming	Opinion: global warming impacts are worsening;...	58.0	58.0

Clean

Missing values in `retweet_count` and `like_count` were replaced using the median value of each column. The median is less sensitive to outliers than the mean and provides a reasonable estimate of central tendency for social media engagement data.

In [3]:
```
# Duplicate detection
print(df.duplicated().sum())
print('----------')

# Find missing values
print(df.isna().sum())
```

```
0
----------
tweet_id          0
date              0
country           0
sentiment         0
keyword           0
text              0
retweet_count    38
like_count       32
dtype: int64
```

In [4]:
```python
# Proper date parsing
df['date'] = pd.to_datetime(df['date'], errors='coerce')

# Duplicate handling
df = df.drop_duplicates()

# Handling missing values in columns with proper justification
df['retweet_count'] = df['retweet_count'].fillna(df['retweet_count'].median())
df['like_count'] = df['like_count'].fillna(df['like_count'].median())

# Display result
df.info()
df
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2000 entries, 0 to 1999
Data columns (total 8 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   tweet_id       2000 non-null   int64
 1   date           2000 non-null   datetime64[ns]
 2   country        2000 non-null   object
 3   sentiment      2000 non-null   object
 4   keyword        2000 non-null   object
 5   text           2000 non-null   object
 6   retweet_count  2000 non-null   float64
 7   like_count     2000 non-null   float64
dtypes: datetime64[ns](1), float64(2), int64(1), object(4)
memory usage: 125.1+ KB
```

Out[4]:

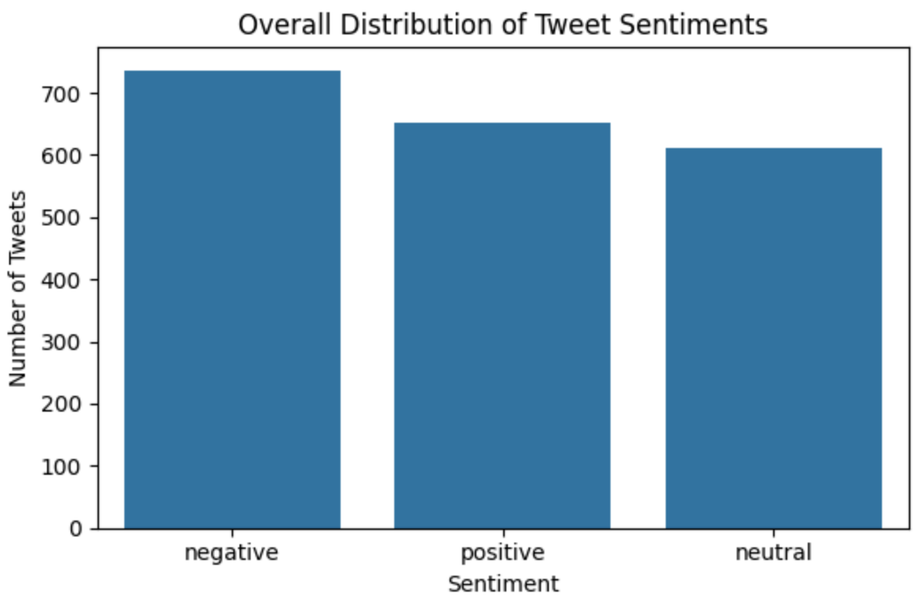
	tweet_id	date	country	sentiment	keyword	text	retweet_count	like_count
0	2803826677020224	2023-11-01	Denmark	negative	air quality	Breaking: concerns grow over air quality; data...	80.0	92.0
1	1119349365897942	2022-11-16	Spain	negative	air quality	Reminder: concerns grow over air quality; deta...	14.0	125.0
2	3605591970800719	2023-01-20	France	neutral	green transition	FYI: discussion continues on green transition;...	78.0	109.0
3	4546715437012377	2023-09-10	Saudi Arabia	negative	green transition	Opinion: concerns grow over green transition; ...	75.0	63.0
4	9630049315224554	2022-09-26	Ireland	negative	global warming	Opinion: global warming impacts are worsening;...	58.0	58.0
...	...	...	...	...	...	...	...	...
1995	6114707426537667	2021-07-14	India	negative	carbon tax	FYI: alarm raised about carbon tax; what do yo...	48.0	110.0
1996	6518489934791567	2021-07-04	China	negative	fossil fuels	Update: concerns grow over fossil fuels; stay ...	23.0	147.0
1997	2244138508075894	2023-11-29	Australia	positive	green transition	Reminder: encouraging signs for green transiti...	69.0	127.0
1998	4958684821663573	2022-04-19	Indonesia	negative	sustainability	Report: alarm raised about sustainability; det...	3.0	65.0
1999	2250898629368768	2021-08-10	Germany	negative	heatwave	Report: alarm raised about heatwave; check the...	48.0	122.0

2000 rows × 8 columns

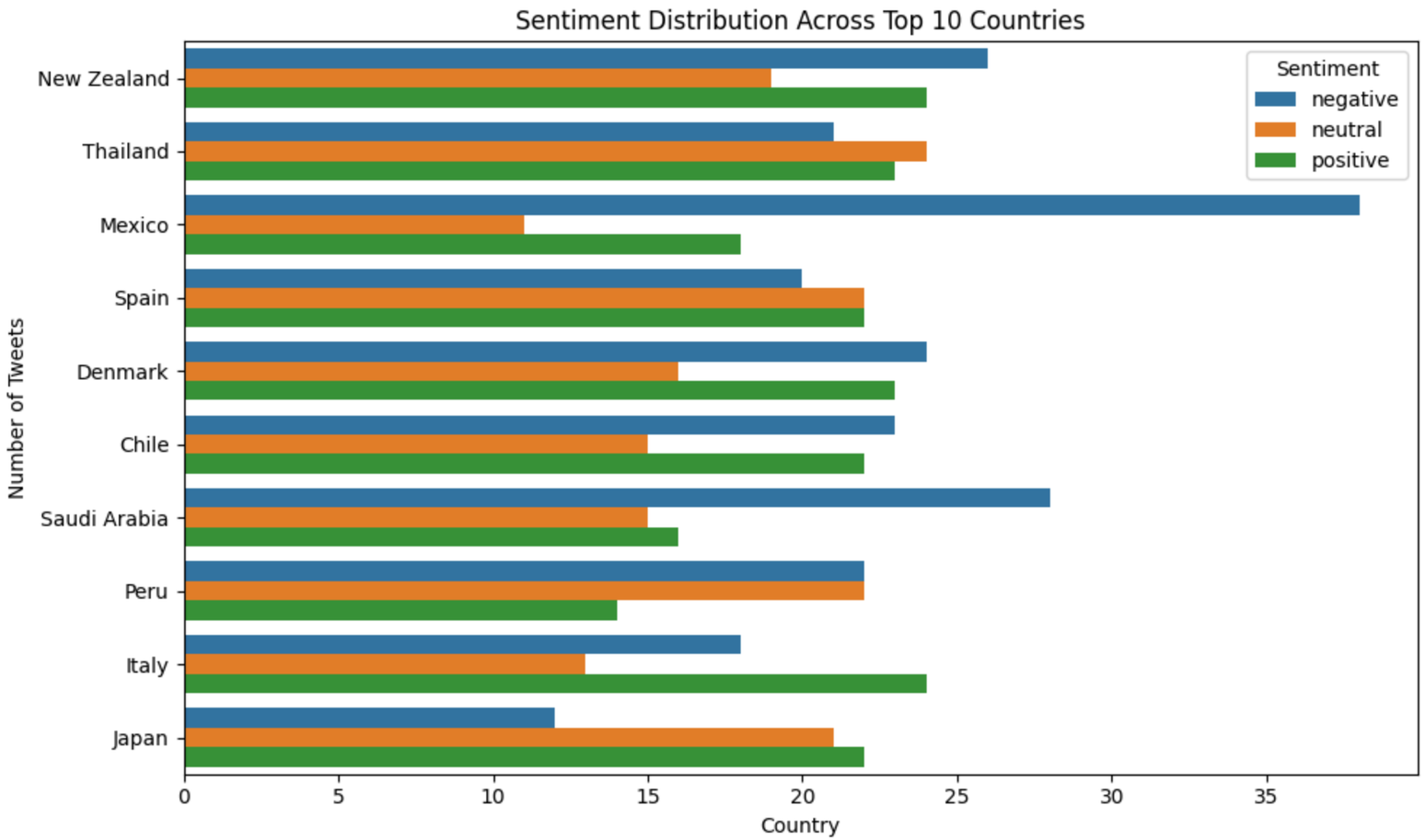
In [5]:
```python
# Lock a cleaned copy for export
df_cleaned = df.copy()
```

Sentiment Visualization

In [6]:
```
plt.figure(figsize=(6,4))

sns.countplot(
    data=df,
    x='sentiment',
    order=df['sentiment'].value_counts().index
)

plt.title("Overall Distribution of Tweet Sentiments")
plt.xlabel("Sentiment")
plt.ylabel("Number of Tweets")

plt.tight_layout()
plt.show()
```



In [7]:
```
top_countries = df['country'].value_counts().head(10).index

plt.figure(figsize=(10,6))

sns.countplot(
    data=df[df['country'].isin(top_countries)],
    y='country',
    hue='sentiment',
    order=top_countries
)

plt.title("Sentiment Distribution Across Top 10 Countries")
plt.xlabel("Country")
plt.ylabel("Number of Tweets")
plt.legend(title="Sentiment")

plt.tight_layout()
plt.show()
```



Part B – Text Preprocess & Word Cloud

In [8]:
```
# Text preprocessing
import re
import nltk
from nltk.corpus import stopwords

# Keyword & word cloud analysis
from collections import Counter
from wordcloud import WordCloud
```

Text Preprocessing

In [9]:
```
nltk.download('stopwords')
stop_words = set(stopwords.words('english'))

def clean_text(text):
    text = text.lower()                     # lowercase
    text = re.sub(r'[^a-z\s]', '', text)    # remove punctuation and numbers
    words = [word for word in text.split() if word not in stop_words]   # remove stopwords

    return " ".join(words)
```

```
df['cleaned_text'] = df['text'].apply(clean_text)
df
```

```
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
```

Out[9]:

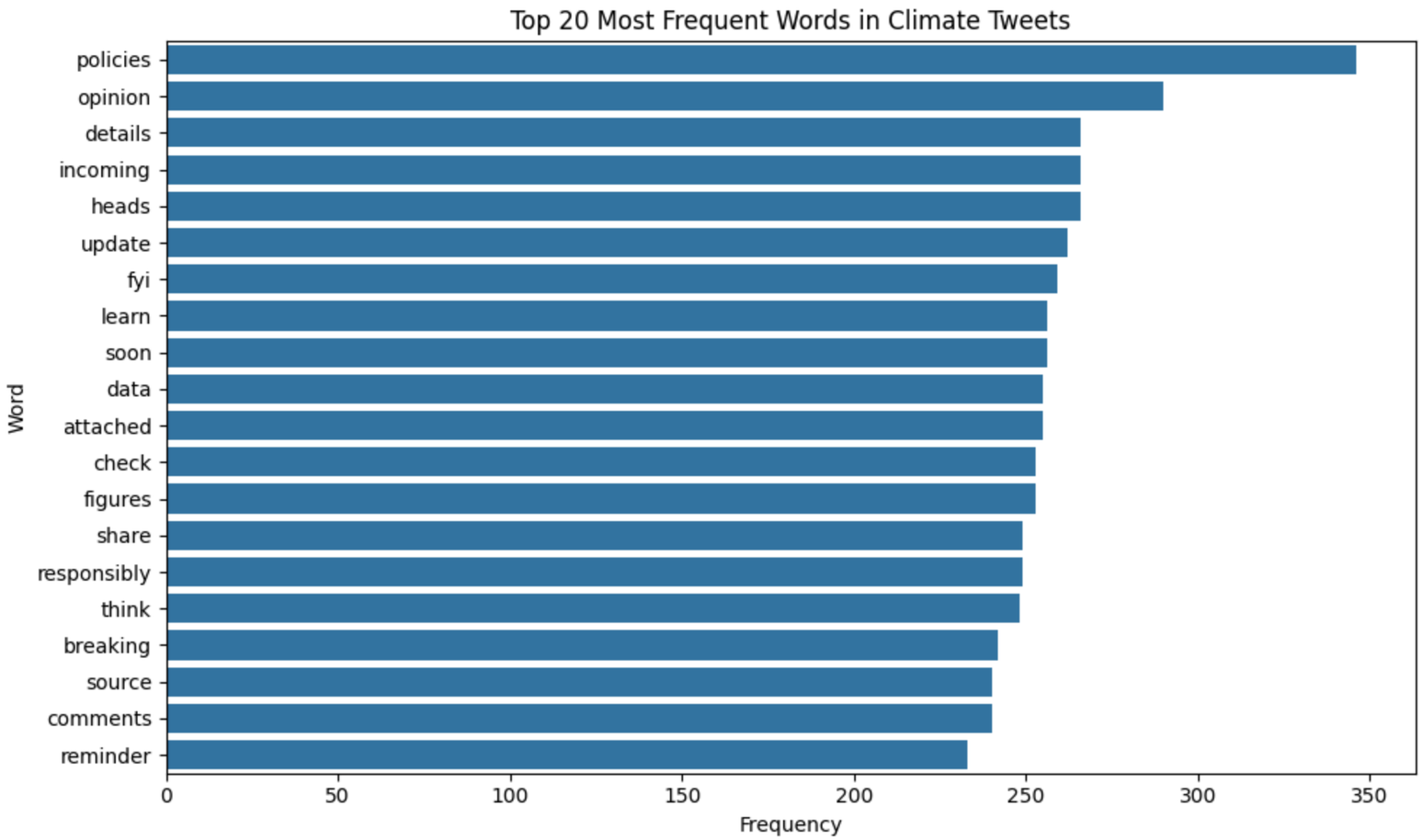
	tweet_id	date	country	sentiment	keyword	text	retweet_count	like_count	cleaned_text
0	2803826677020224	2023-11-01	Denmark	negative	air quality	Breaking: concerns grow over air quality; data...	80.0	92.0	breaking concerns grow air quality data attached
1	1119349365897942	2022-11-16	Spain	negative	air quality	Reminder: concerns grow over air quality; deta...	14.0	125.0	reminder concerns grow air quality details inc...
2	3605591970800719	2023-01-20	France	neutral	green transition	FYI: discussion continues on green transition;...	78.0	109.0	fyi discussion continues green transition chec...
3	4546715437012377	2023-09-10	Saudi Arabia	negative	green transition	Opinion: concerns grow over green transition; ...	75.0	63.0	opinion concerns grow green transition check f...
4	9630049315224554	2022-09-26	Ireland	negative	global warming	Opinion: global warming impacts are worsening;...	58.0	58.0	opinion global warming impacts worsening stay ...
...	...	...	...	...	...	...	...	...	...
1995	6114707426537667	2021-07-14	India	negative	carbon tax	FYI: alarm raised about carbon tax; what do yo...	48.0	110.0	fyi alarm raised carbon tax think
1996	6518489934791567	2021-07-04	China	negative	fossil fuels	Update: concerns grow over fossil fuels; stay ...	23.0	147.0	update concerns grow fossil fuels stay informed
1997	2244138508075894	2023-11-29	Australia	positive	green transition	Reminder: encouraging signs for green transiti...	69.0	127.0	reminder encouraging signs green transition st...
1998	4958684821663573	2022-04-19	Indonesia	negative	sustainability	Report: alarm raised about sustainability; det...	3.0	65.0	report alarm raised sustainability details inc...
1999	2250898629368768	2021-08-10	Germany	negative	heatwave	Report: alarm raised about heatwave; check the...	48.0	122.0	report alarm raised heatwave check figures

2000 rows × 9 columns

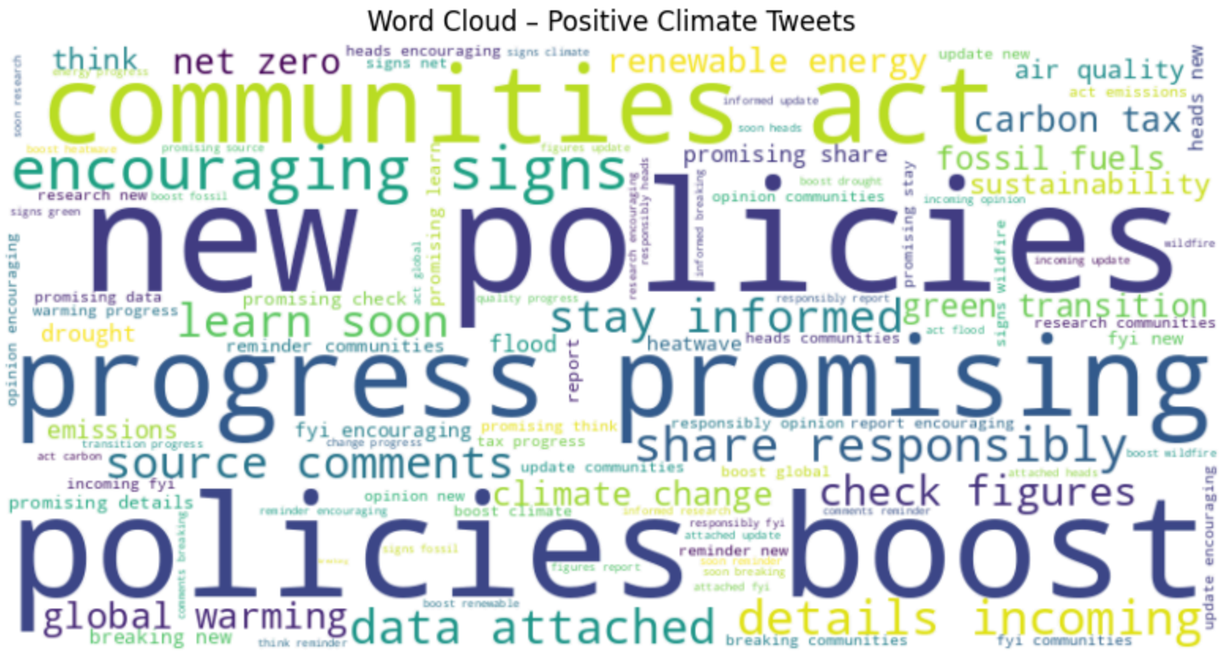
Keyword & Word Cloud Analysis

In [10]:

```
all_words = " ".join(df['cleaned_text']).split()    # Tokenize words
top20 = Counter(all_words).most_common(20)          # Count frequency
freq_df = pd.DataFrame(top20, columns=['word', 'count'])    # to DataFrame

plt.figure(figsize=(10,6))

sns.barplot(
    data=freq_df,
    x='count',
    y='word'
)

plt.title("Top 20 Most Frequent Words in Climate Tweets")
plt.xlabel("Frequency")
plt.ylabel("Word")

plt.tight_layout()
plt.show()
```

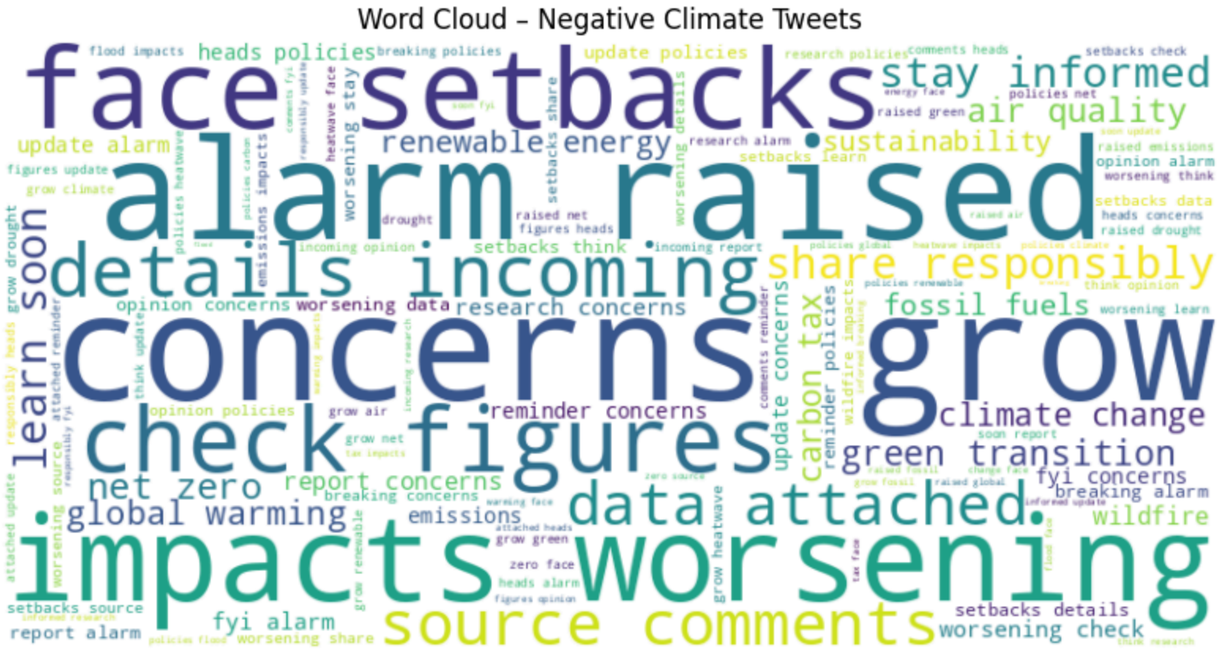


In [11]:
```
positive_text = " ".join(df[df["sentiment"] == "positive"]["cleaned_text"])

wordcloud_pos = WordCloud(
    width=800,
    height=400,
    background_color="white"
).generate(positive_text)

plt.figure(figsize=(10,5))
plt.imshow(wordcloud_pos, interpolation="bilinear")
plt.axis("off")

plt.title("Word Cloud – Positive Climate Tweets")

plt.show()
```



In [12]:

```
negative_text = " ".join(df[df["sentiment"] == "negative"]["cleaned_text"])

wordcloud_neg = WordCloud(
    width=800,
    height=400,
    background_color="white"
).generate(negative_text)

plt.figure(figsize=(10,5))
plt.imshow(wordcloud_neg, interpolation="bilinear")
plt.axis("off")

plt.title("Word Cloud – Negative Climate Tweets")

plt.show()
```



Part C – Geospatial Visualization

In [13]:
```
!pip install --upgrade plotly
!pip install pycountry kaleido

import pycountry
import plotly.express as px
from IPython.core.display import display, HTML
```

```
Requirement already satisfied: plotly in /usr/local/lib/python3.12/dist-packages (6.6.0)
Requirement already satisfied: narwhals>=1.15.1 in /usr/local/lib/python3.12/dist-packages (from plotly) (2.17.0)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from plotly) (26.0)
Requirement already satisfied: pycountry in /usr/local/lib/python3.12/dist-packages (26.2.16)
Requirement already satisfied: kaleido in /usr/local/lib/python3.12/dist-packages (1.2.0)
Requirement already satisfied: choreographer>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from kaleido) (1.2.1)
Requirement already satisfied: logistro>=1.0.8 in /usr/local/lib/python3.12/dist-packages (from kaleido) (2.0.1)
Requirement already satisfied: orjson>=3.10.15 in /usr/local/lib/python3.12/dist-packages (from kaleido) (3.11.7)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from kaleido) (26.0)
Requirement already satisfied: pytest-timeout>=2.4.0 in /usr/local/lib/python3.12/dist-packages (from kaleido) (2.4.0)
Requirement already satisfied: simplejson>=3.19.3 in /usr/local/lib/python3.12/dist-packages (from choreographer>=1.1.1->kaleido) (3.20.2)
Requirement already satisfied: pytest>=7.0.0 in /usr/local/lib/python3.12/dist-packages (from pytest-timeout>=2.4.0->kaleido) (8.4.2)
Requirement already satisfied: iniconfig>=1 in /usr/local/lib/python3.12/dist-packages (from pytest>=7.0.0->pytest-timeout>=2.4.0->kaleido) (2.3.0)
Requirement already satisfied: pluggy<2,>=1.5 in /usr/local/lib/python3.12/dist-packages (from pytest>=7.0.0->pytest-timeout>=2.4.0->kaleido) (1.6.0)
Requirement already satisfied: pygments>=2.7.2 in /usr/local/lib/python3.12/dist-packages (from pytest>=7.0.0->pytest-timeout>=2.4.0->kaleido) (2.19.2)
```

In [14]:
```
# Convert to ISO-3 country codes (stop deprecation warning)
country_fix = {     # manual name fix
    "Russia": "Russian Federation",
    "Turkey": "Türkiye"
}

def get_iso3(country_name):
    try:
        name_fixed = country_fix.get(country_name, country_name)
```

```
        return pycountry.countries.lookup(name_fixed).alpha_3
    except LookupError:
        return None

df['iso_country'] = df['country'].apply(get_iso3)

# Check for failed conversions
df[df['iso_country'].isna()]['country'].unique()
```

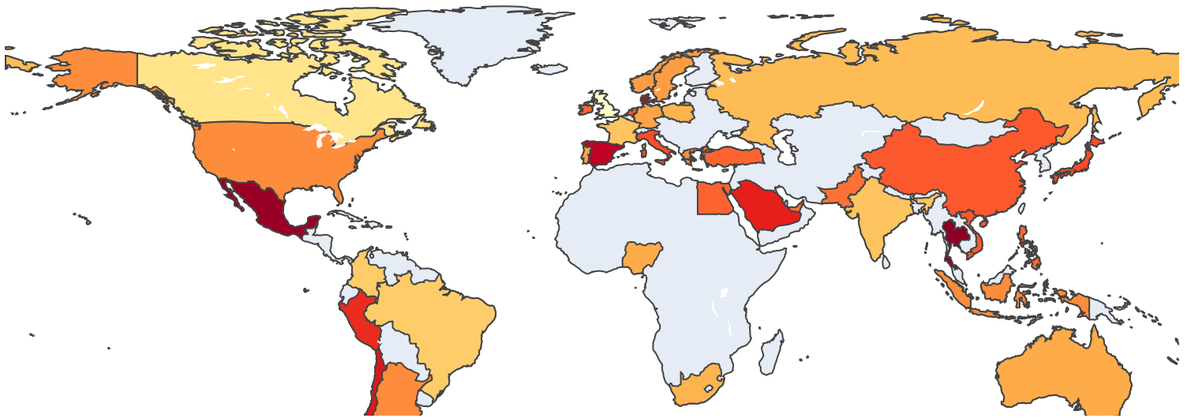
Out[14]:

```
array([], dtype=object)
```

In [15]:

```
country_counts = df.groupby('iso_country').size().reset_index(name="tweet_count")

fig = px.choropleth(
    country_counts,
    locations="iso_country",        # column with country names
    locationmode="ISO-3",
    color="tweet_count",        # metric to visualize
    hover_name="iso_country",
    color_continuous_scale="YlOrRd",  # Yellow → Orange → Red
)

fig.update_layout(
    title_text="Number of Climate Tweets by Country",
    geo=dict(showframe=False, showcoastlines=True),
)

fig.show()  # html data, does not render on notebook

# fig.write_html("climate_tweets_map.html")
# with open("climate_tweets_map.html", "r") as f:
#     html_content = f.read()
# display(HTML(html_content))
```

Number of Climate Tweets by Country



Part D – Uncertainty & Association

Monthly Negative Sentiment Rate + 95% confidence interval (CI).

To compute the monthly negative sentiment rate, I divided the number of negative tweets by the total tweets each month and calculated 95% confidence intervals using the Wilson method, which is suitable for proportions and accounts for uncertainty in months with fewer tweets. The plot shows that negative sentiment fluctuates over time, with some months exhibiting higher public concern about climate issues. The shaded CI highlights uncertainty, indicating when the rate estimates are less precise due to smaller sample sizes.

In [16]:
```
from statsmodels.stats.proportion import proportion_confint
```

In [17]:
```
# Process dataset
df['month'] = df['date'].dt.to_period('M')
monthly_counts = df.groupby('month')['sentiment'].value_counts().unstack(fill_value=0)
monthly_counts['total'] = monthly_counts.sum(axis=1)
monthly_counts['neg_rate'] = monthly_counts.get('negative', 0) / monthly_counts['total']

# Compute 95% CI using Wilson method
confint = monthly_counts.apply(
    lambda row: proportion_confint(
        count=row.get('negative', 0),
        nobs=row['total'],
        alpha=0.05,
        method='wilson'
    ), axis=1
)
```

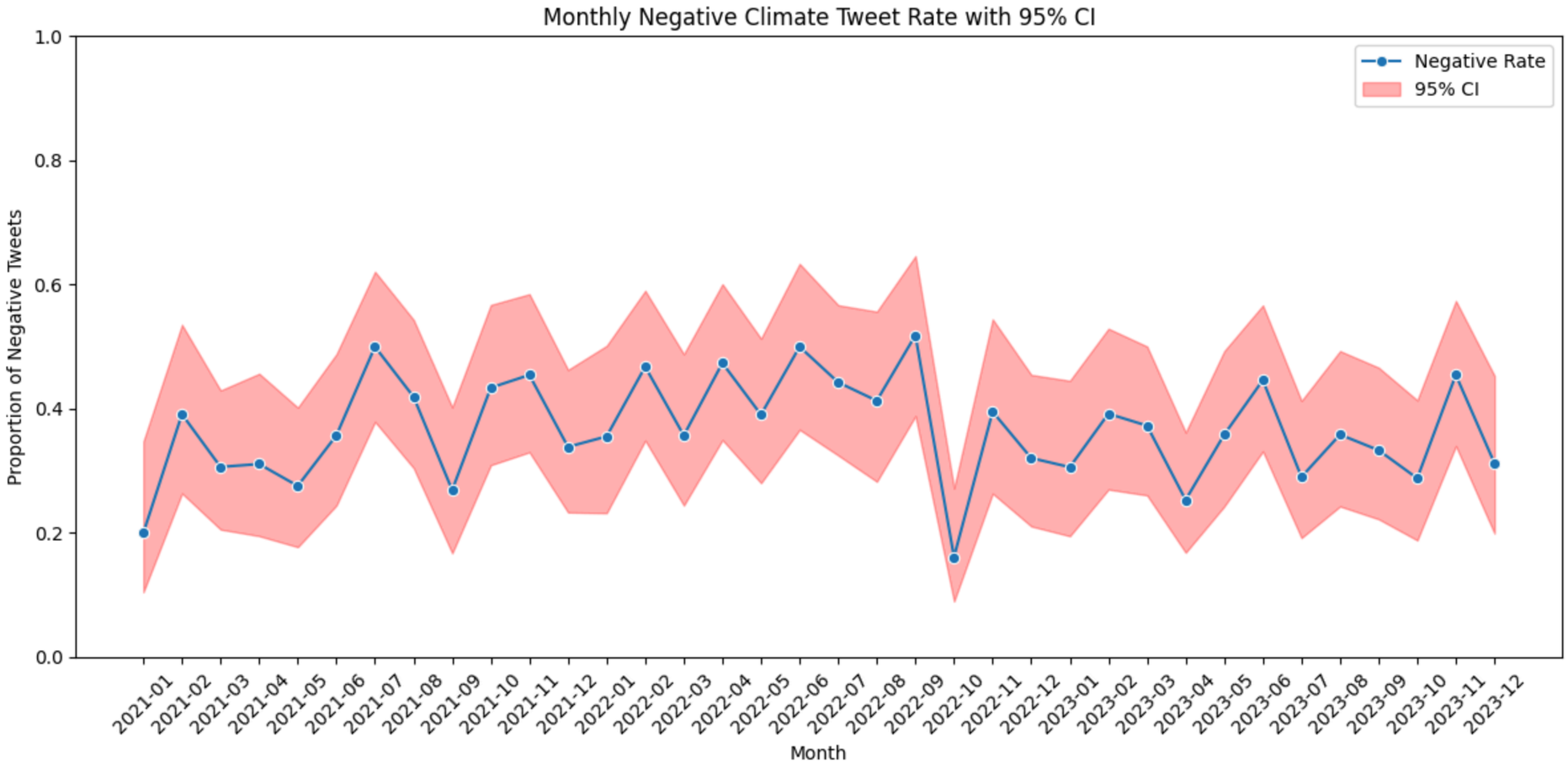
```
monthly_counts['ci_lower'] = [c[0] for c in confint]
monthly_counts['ci_upper'] = [c[1] for c in confint]

monthly_counts.head()
```

Out[17]:

sentiment	negative	neutral	positive	total	neg_rate	ci_lower	ci_upper
month							
2021-01	8	15	17	40	0.200000	0.105000	0.347573
2021-02	18	17	11	46	0.391304	0.263932	0.535432
2021-03	19	21	22	62	0.306452	0.205816	0.429672
2021-04	14	16	15	45	0.311111	0.195287	0.456648
2021-05	16	18	24	58	0.275862	0.177524	0.402047

In [18]:

```
plt.figure(figsize=(12,6))

sns.lineplot(
    x=monthly_counts.index.astype(str),
    y=monthly_counts['neg_rate'],
    marker='o',
    label='Negative Rate'
)

plt.fill_between(
    monthly_counts.index.astype(str),
    monthly_counts['ci_lower'],
    monthly_counts['ci_upper'],
    alpha=0.3, color='red', label='95% CI'
)

plt.title("Monthly Negative Climate Tweet Rate with 95% CI")
plt.xlabel("Month")
plt.ylabel("Proportion of Negative Tweets")
plt.xticks(rotation=45)
plt.legend()
plt.ylim(0, 1)

plt.tight_layout()
plt.show()
```

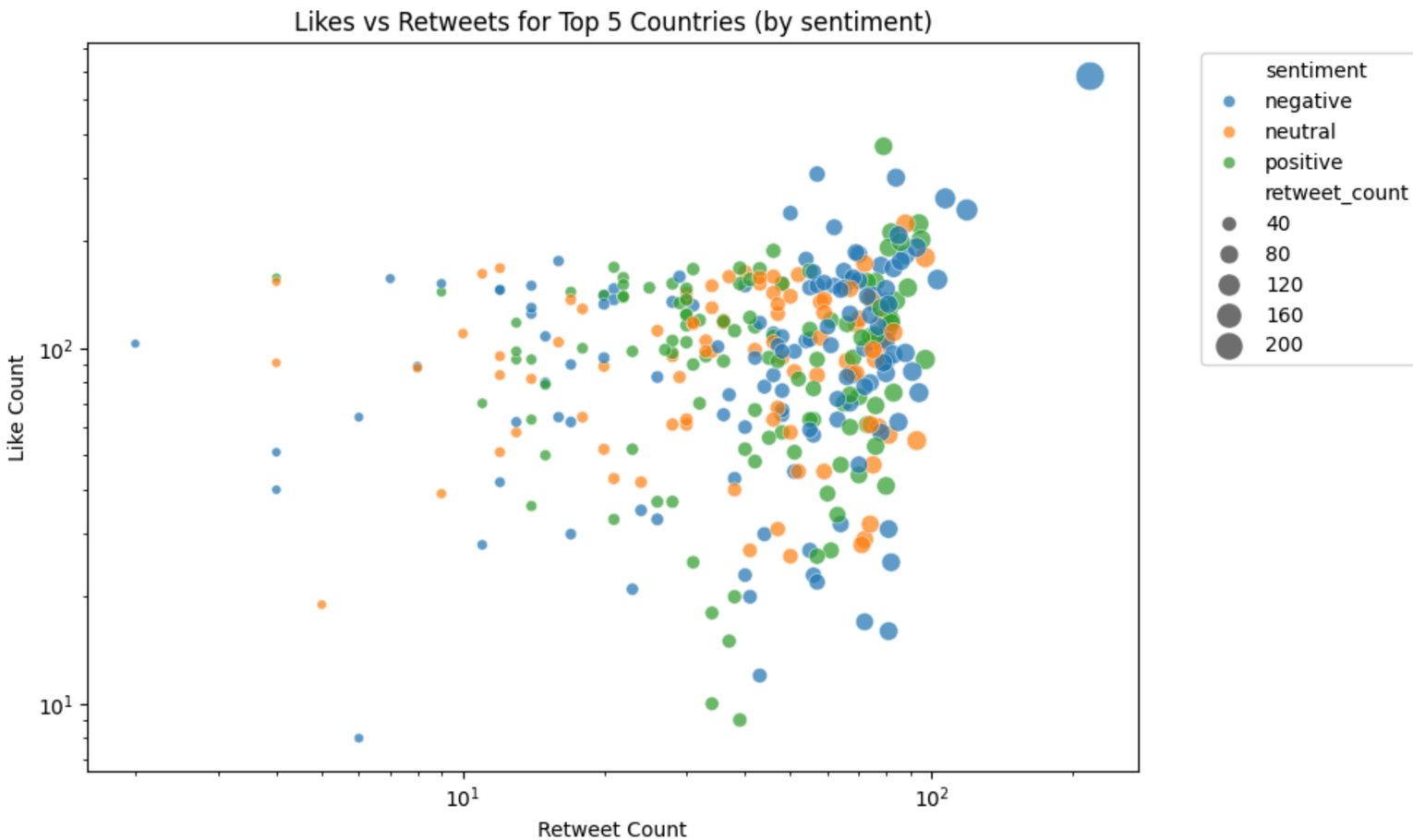


Scatter/Bubble: Likes vs Retweets

I plotted likes versus retweets for the top 5 countries to avoid overcrowding and used color to represent sentiment and bubble size to indicate retweet magnitude. The scatter plot shows a positive correlation between likes and retweets, with some negative tweets receiving particularly high engagement. This visualization helps identify which types of tweets attract attention and allows comparison of engagement patterns across sentiments and countries.

In [19]:
```
top_countries = df['country'].value_counts().head(5).index
df_top = df[df['country'].isin(top_countries)]

plt.figure(figsize=(10,6))

sns.scatterplot(
    data=df_top,
    x='retweet_count',
    y='like_count',
    hue='sentiment',
    size='retweet_count',
    sizes=(20, 200),
    alpha=0.7
)

plt.title("Likes vs Retweets for Top 5 Countries (by sentiment)")
plt.xlabel("Retweet Count")
plt.ylabel("Like Count")
plt.xscale('log')
```

```
plt.yscale('log')
plt.legend(bbox_to_anchor=(1.05,1), loc='upper left')

plt.tight_layout()
plt.show()
```



Part E – Design & Insight

Design Choices: I used consistent color coding across visualizations to represent sentiment: red for negative, green for positive, and gray/orange for neutral, which helps quickly identify patterns. Bar charts and scatter plots have clear axis labels and legends, while the choropleth uses a sequential color scale to emphasize higher tweet counts per country. For the monthly negative sentiment rate, the Wilson 95% confidence interval is shown as a shaded area, visually communicating uncertainty in months with fewer tweets. Log scales were applied to the scatter plot to handle skewed distributions in likes and retweets, improving readability.

Insights: The analysis shows that negative climate tweets are consistently present, but certain months experience spikes in public concern. Some countries dominate the conversation, particularly those with larger tweet counts, and the choropleth highlights geographic concentration of climate discourse. The scatter plot indicates that high-engagement tweets are not limited to positive or neutral sentiment; negative tweets often generate considerable interaction, suggesting that concern or alarm drives sharing. Word cloud and keyword analysis revealed recurring topics like air quality, carbon tax, and green transition, reflecting the issues that capture public attention. Overall, the visualizations collectively reveal temporal, geographic, and sentiment-driven patterns in climate-related discourse, offering both a quantitative and qualitative understanding of public engagement.

Export

In [20]:
```
# Export cleaned CSV
df_cleaned.to_csv("cleaned.csv", index=False)
```